

Visualizing Spatial and Spectral Differences in EEG During Emotion Processing Using Spectrograms

Some Background:

Different patterns of neural activity are associated with different frequency bands:

- **Delta:** approximately 0.5–4 Hz
- **Theta:** approximately 4–8 Hz
- **Alpha:** approximately 8–13 Hz
- **Beta:** approximately 13–30 Hz

In an emotion-related task whether EEG activity differs across brain regions and whether these differences are reflected in particular frequency bands.

The central question is: “The central question of this laboratory is: Do different brain regions exhibit different spectral patterns during emotional processing, and what might those differences tell us about the neural processing of emotion? “

You should investigate questions such as:

- Do frontal, central, temporal, parietal, and occipital electrodes exhibit different spectral patterns?
- Which frequency bands show the strongest activity at different locations?
- Does alpha activity differ between anterior and posterior regions?
- Are theta or beta-band activities particularly prominent at particular locations?
- Do spectral patterns change during emotionally salient portions of the stimulus?
- Can spatial–spectral differences provide clues about the distributed nature of emotion processing?

Dataset: DEAP Dataset (<https://www.kaggle.com/datasets/manh123df/deap-dataset>)

Supporting Paper: EEG Based Emotion Recognition: A Tutorial and Review